

МИНОБРНАУКИ РОССИИ

Федеральное государственное бюджетное образовательное учреждение высшего образования



НИЖЕГОРОДСКИЙ ГОСУДАРСТВЕННЫЙ ТЕХНИЧЕСКИЙ
УНИВЕРСИТЕТ им. Р.Е.АЛЕКСЕЕВА

Институт радиоэлектроники и информационных технологий
Кафедра информатики и систем управления

Лабораторная работа № 3 «Система контроля версий GIT»
(наименование темы проекта или работы)

ОТЧЁТ

По лабораторной работе
по дисциплине

Информатика и компьютерные технологии
(наименование дисциплины)

РУКОВОДИТЕЛЬ:

(подпись)

Савкин А.Е.
(фамилия, и.,о.)

СТУДЕНТ:

(подпись)

Козлов И.С.
(фамилия, и.,о.)

25-ИСз-1
(шифр группы)

Работа защищена «__» _____

С оценкой _____

Нижний Новгород 2026

Задание к выполнению лабораторной работы № 3

Реализация пошаговых блок-схем алгоритмов

Вариант 22

Задание на лабораторную работу:

Загрузить отчеты и материалы по лабораторным работам в репозиторий на GitHub.

Цель работы:

Научиться работать с системой контроля версий GIT.

Ход работы:

Создал локальный репозиторий

```
vboxuser@10:~/Documents$ cd Repo/
vboxuser@10:~/Documents/Repo$ git clone https://github.com/kozlov-ilya11/Spring_2026.git
Cloning into 'Spring_2026'...
remote: Enumerating objects: 20, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 20 (delta 0), reused 0 (delta 0), pack-reused 16 (from 1)
Receiving objects: 100% (20/20), 35.39 MiB | 607.00 KiB/s, done.
vboxuser@10:~/Documents/Repo$ ls
Spring_2026
vboxuser@10:~/Documents/Repo$ cd Spring_2026/
vboxuser@10:~/Documents/Repo/Spring_2026$ git branch Kozlov_I_S_25ISz
vboxuser@10:~/Documents/Repo/Spring_2026$ git branch
  Kozlov_I_S_25ISz
* main
vboxuser@10:~/Documents/Repo/Spring_2026$
```

Сделал первый commit

```
vboxuser@10:~/Documents/Repo/Spring_2026$ git add .
vboxuser@10:~/Documents/Repo/Spring_2026$ git commit -m "Test commit"
[Kozlov_I_S_25ISz ff6b5a5] Test commit
3 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 Lr1/test.txt
create mode 100644 Lr2/test.txt
create mode 100644 Lr3/test.txt
vboxuser@10:~/Documents/Repo/Spring_2026$ git branch
* Kozlov_I_S_25ISz
  main
vboxuser@10:~/Documents/Repo/Spring_2026$ nano Lr1/test.txt
vboxuser@10:~/Documents/Repo/Spring_2026$ nano Lr2/test.txt
vboxuser@10:~/Documents/Repo/Spring_2026$ nano Lr3/test.txt
vboxuser@10:~/Documents/Repo/Spring_2026$
```

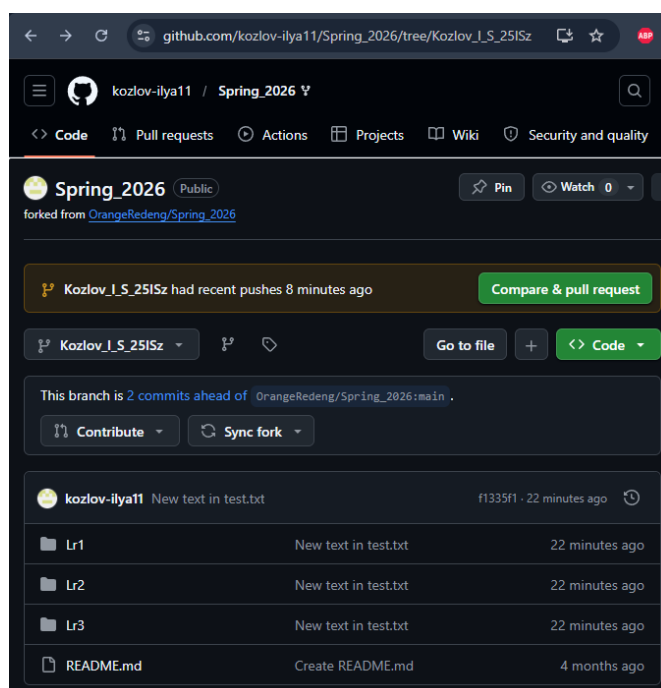
Добавил текст в файлы test.txt и сделал commit

```
vboxuser@10:~/Documents/Repo/Spring_2026$ git commit -a -m "New text in test.txt"
[Kozlov_I_S_25ISz f1335f1] New text in test.txt
 3 files changed, 3 insertions(+)
vboxuser@10:~/Documents/Repo/Spring_2026$ nano Lr3/test.txt
vboxuser@10:~/Documents/Repo/Spring_2026$ git commit -m "New text in Lr1/test.txt"
t"
On branch Kozlov_I_S_25ISz
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Lr3/test.txt

no changes added to commit (use "git add" and/or "git commit -a")
vboxuser@10:~/Documents/Repo/Spring_2026$
```

Добавил изменения в общий репозиторий

```
vboxuser@10:~/Documents/Repo/Spring_2026$ git push --set-upstream origin Kozlov_I_S_25ISz
Username for 'https://github.com': kozlov-ilya11
Password for 'https://kozlov-ilya11@github.com':
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Compressing objects: 100% (10/10), done.
Writing objects: 100% (12/12), 1.06 KiB | 1.06 MiB/s, done.
Total 12 (delta 6), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (6/6), completed with 3 local objects.
remote:
remote: Create a pull request for 'Kozlov_I_S_25ISz' on GitHub by visiting:
remote:   https://github.com/kozlov-ilya11/Spring_2026/pull/new/Kozlov_I_S_25ISz
remote:
To https://github.com/kozlov-ilya11/Spring_2026.git
 * [new branch]      Kozlov_I_S_25ISz -> Kozlov_I_S_25ISz
branch 'Kozlov_I_S_25ISz' set up to track 'origin/Kozlov_I_S_25ISz'.
vboxuser@10:~/Documents/Repo/Spring_2026$
```

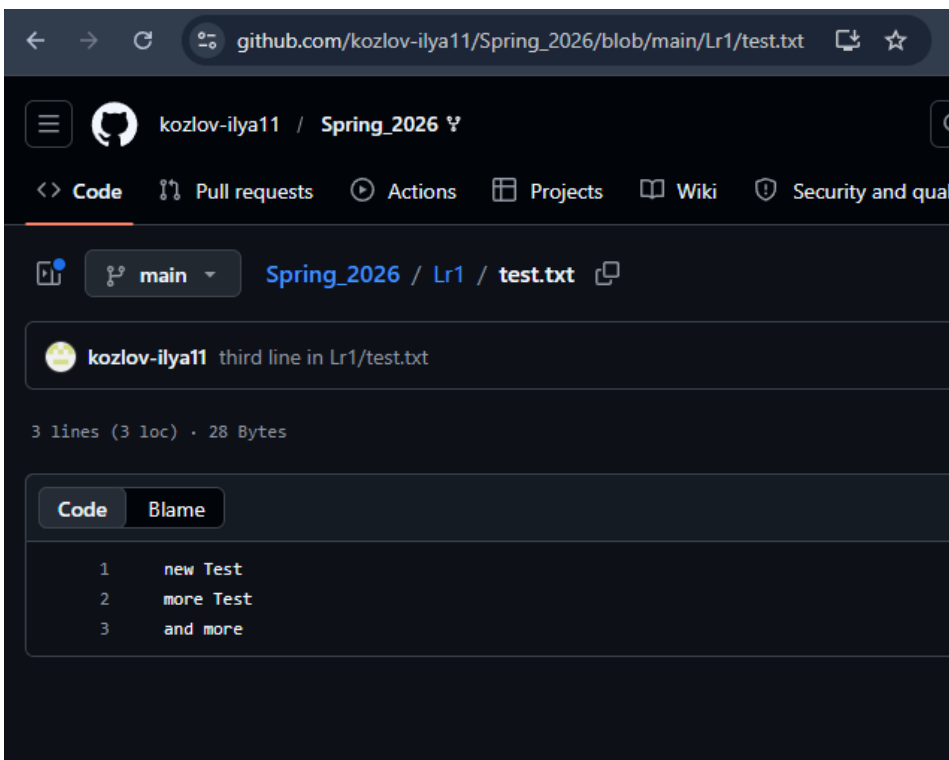


Добавил в text.txt ещё одну строку и выполнил push в общий репозиторий

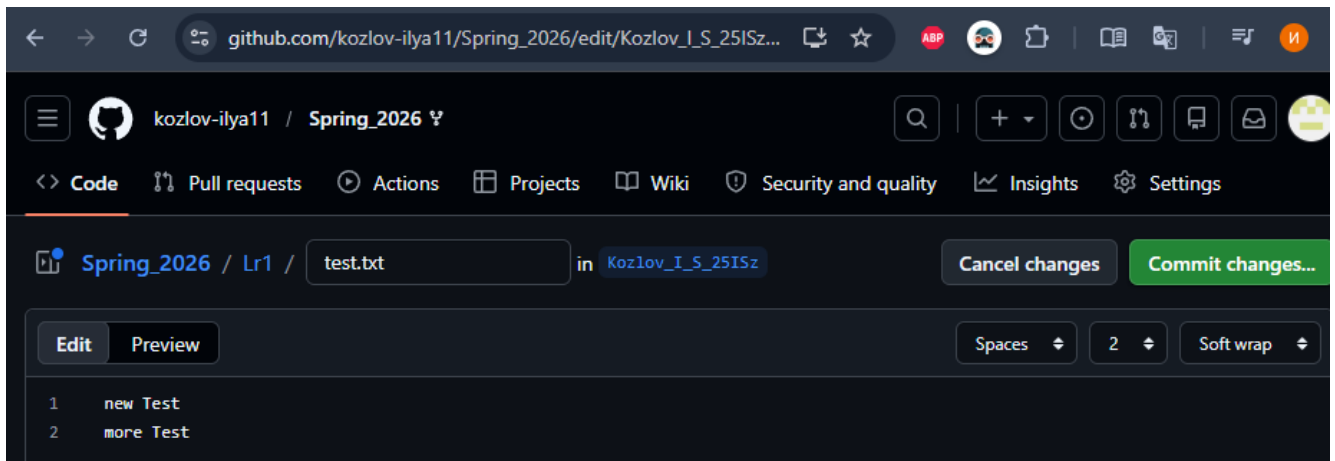
```
vboxuser@10:~/Documents/Repo/Spring_2026$ ls Lr1
test.txt  Блок-Схемы.pptx  'ИиКТ_Лаб1 2026.doc'  'Пример отчёта.docx'
vboxuser@10:~/Documents/Repo/Spring_2026$ git add .
vboxuser@10:~/Documents/Repo/Spring_2026$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Lr1/test.txt

vboxuser@10:~/Documents/Repo/Spring_2026$ git commit -m -a "third line in Lr1/test.txt"
error: pathspec 'third line in Lr1/test.txt' did not match any file(s) known to git
vboxuser@10:~/Documents/Repo/Spring_2026$ git commit -m "third line in Lr1/test.txt"
[main 0bf5bac] third line in Lr1/test.txt
1 file changed, 3 insertions(+)
create mode 100644 Lr1/test.txt
vboxuser@10:~/Documents/Repo/Spring_2026$ git push --set-upstream origin main
Username for 'https://github.com': kozlov-ilya11
Password for 'https://kozlov-ilya11@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 433 bytes | 433.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/kozlov-ilya11/Spring_2026.git
66c3c1b..0bf5bac  main -> main
branch 'main' set up to track 'origin/main'.
vboxuser@10:~/Documents/Repo/Spring_2026$
```

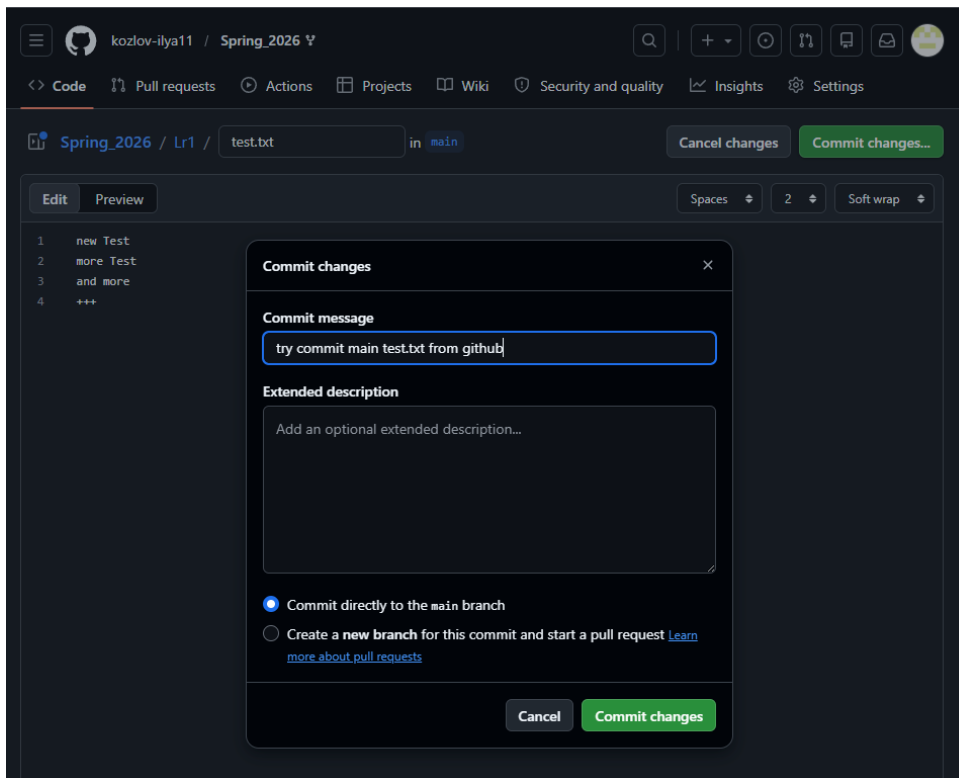


Добавил строку в test.txt общего репозитория через GitHub и сделал pull в локальный репозиторий



```
vboxuser@10:~/Documents/Repo/Spring_2026$ git branch
* Kozlov_I_S_25ISz
  main
vboxuser@10:~/Documents/Repo/Spring_2026$ git pull
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.04 KiB | 1.04 MiB/s, done.
From https://github.com/kozlov-ilya11/Spring_2026
   f1335f1..dcfa900  Kozlov_I_S_25ISz -> origin/Kozlov_I_S_25ISz
Updating f1335f1..dcfa900
Fast-forward
 Lr1/test.txt | 1 +
 1 file changed, 1 insertion(+)
vboxuser@10:~/Documents/Repo/Spring_2026$ ls
Lr1  Lr2  Lr3  README.md
vboxuser@10:~/Documents/Repo/Spring_2026$ cat Lr1/test.txt
new Test
more Test
```

Добавил строку в test.txt общего репозитория через GitHub и сделал pull в локальный репозиторий

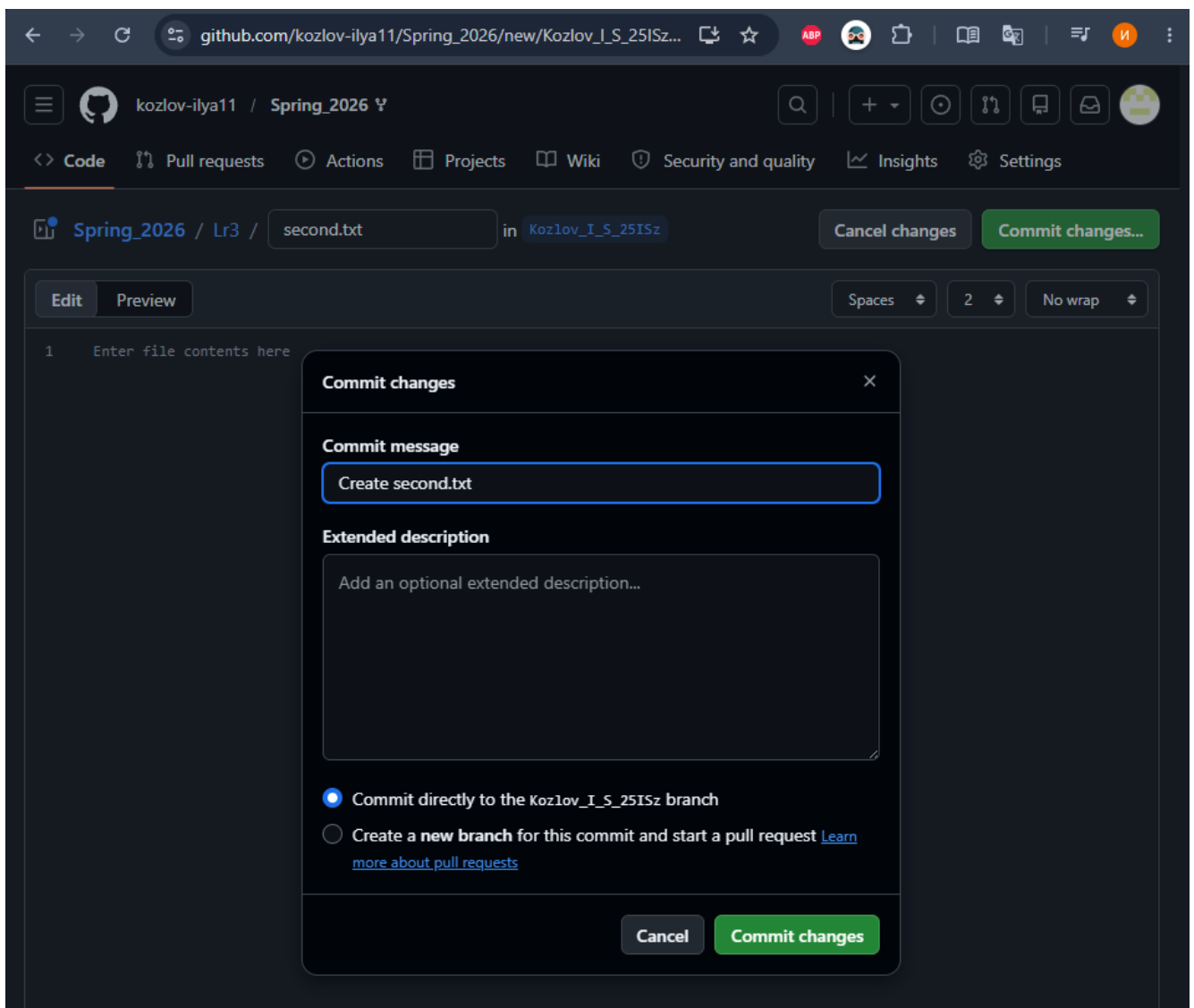


```
vboxuser@10:~/Documents/Repo/Spring_2026$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
vboxuser@10:~/Documents/Repo/Spring_2026$ git branch
Kozlov_I_S_25ISz
* main
vboxuser@10:~/Documents/Repo/Spring_2026$ ls Lr1/test.txt
Lr1/test.txt
vboxuser@10:~/Documents/Repo/Spring_2026$ cat Lr1/test.txt
new Test
more Test
and more
vboxuser@10:~/Documents/Repo/Spring_2026$ git pull
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.06 KiB | 1.06 MiB/s, done.
From https://github.com/kozlov-ilya11/Spring_2026
   0bf5bac..a4df7c9  main       -> origin/main
Updating 0bf5bac..a4df7c9
Fast-forward
 Lr1/test.txt | 1 +
 1 file changed, 1 insertion(+)
vboxuser@10:~/Documents/Repo/Spring_2026$ cat Lr1/test.txt
new Test
more Test
and more
+++
vboxuser@10:~/Documents/Repo/Spring_2026$
```

Изменил файл в локальной репозитории

```
vboxuser@10: ~/Documents/Repo/Spring_2026$ git branch
* Kozlov_I_S_25ISz
  main
vboxuser@10: ~/Documents/Repo/Spring_2026$ nano Lr3/test.txt
vboxuser@10: ~/Documents/Repo/Spring_2026$ cat Lr3/test.txt
new Test
text2
vboxuser@10: ~/Documents/Repo/Spring_2026$
```

Добавил файл в общем репозитории



Конфликт при выполнении push

```
vboxuser@10:~/Documents/Repo/Spring_2026$ git push --set-upstream origin Kozlov_I_S_25ISz
Username for 'https://github.com': kozlov-ilya11
Password for 'https://kozlov-ilya11@github.com':
To https://github.com/kozlov-ilya11/Spring_2026.git
 ! [rejected]        Kozlov_I_S_25ISz -> Kozlov_I_S_25ISz (fetch first)
error: failed to push some refs to 'https://github.com/kozlov-ilya11/Spring_2026.git'
hint: Updates were rejected because the remote contains work that you do
hint: not have locally. This is usually caused by another repository pushing
hint: to the same ref. You may want to first integrate the remote changes
hint: (e.g., 'git pull ...') before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
vboxuser@10:~/Documents/Repo/Spring_2026$
```

Выполнил pull для устранения конфликта. Push выполнен успешно

```
vboxuser@10:~/Documents/Repo/Spring_2026$ git pull
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 1), reused 1 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.02 KiB | 348.00 KiB/s, done.
From https://github.com/kozlov-ilya11/Spring_2026
   dcfa900..4fc2408  Kozlov_I_S_25ISz -> origin/Kozlov_I_S_25ISz
Updating dcfa900..4fc2408
Fast-forward
 Lr3/second.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 Lr3/second.txt
vboxuser@10:~/Documents/Repo/Spring_2026$ cat Lr3/second.txt

vboxuser@10:~/Documents/Repo/Spring_2026$ ls Lr3
second.txt  'ИиКТ_Лаб_Git_v3.4 2026.doc'
test.txt    'Системы контроля версий_v3.4 2026.pptx'
vboxuser@10:~/Documents/Repo/Spring_2026$ cat Lr3/test.txt
new Test
text2
vboxuser@10:~/Documents/Repo/Spring_2026$ git push --set-upstream origin Kozlov_I_S_25ISz
Username for 'https://github.com': kozlov-ilya11
Password for 'https://kozlov-ilya11@github.com':
branch 'Kozlov_I_S_25ISz' set up to track 'origin/Kozlov_I_S_25ISz'.
Everything up-to-date
vboxuser@10:~/Documents/Repo/Spring_2026$
```


Добавил лабораторные в общий репозиторий

```
vboxuser@10:~/Documents/Repo/Spring_2026$ git commit -m "Add lab 1 and 2"
[Kozlov_I_S_25ISz dc6e8a4] Add lab 1 and 2
6 files changed, 1 insertion(+)
create mode 100755 "Lr1/\320\230\320\270\320\232\320\242_25_\320\230\320\241\320\267_\320\23
2\320\276\320\267\320\273\320\276\320\262_\320\230\320\273\321\214\321\217_\320\241\320\265\3
21\200\320\263\320\265\320\265\320\262\320\270\321\207_\320\233\320\240_1.docx"
create mode 100755 "Lr1/\320\230\320\270\320\232\320\242_25_\320\230\320\241\320\267_\320\23
2\320\276\320\267\320\273\320\276\320\262_\320\230\320\273\321\214\321\217_\320\241\320\265\3
21\200\320\263\320\265\320\265\320\262\320\270\321\207_\320\233\320\240_1.pdf"
create mode 100755 Lr2/SITE.zip
create mode 100755 "Lr2/\320\230\320\270\320\232\320\242_25_\320\230\320\241\320\267_\320\23
2\320\276\320\267\320\273\320\276\320\262_\320\230\320\273\321\214\321\217_\320\241\320\265\3
21\200\320\263\320\265\320\265\320\262\320\270\321\207_\320\233\320\240_2.docx"
create mode 100755 "Lr2/\320\230\320\270\320\232\320\242_25_\320\230\320\241\320\267_\320\23
2\320\276\320\267\320\273\320\276\320\262_\320\230\320\273\321\214\321\217_\320\241\320\265\3
21\200\320\263\320\265\320\265\320\262\320\270\321\207_\320\233\320\240_2.pdf"
vboxuser@10:~/Documents/Repo/Spring_2026$ git push --set-upstream origin Kozlov_I_S_25ISz
Username for 'https://github.com': kozlov-ilya11
Password for 'https://kozlov-ilya11@github.com':
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Compressing objects: 100% (10/10), done.
Writing objects: 100% (11/11), 6.65 MiB | 1.04 MiB/s, done.
Total 11 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/kozlov-ilya11/Spring_2026.git
 4fc2408..dc6e8a4 Kozlov_I_S_25ISz -> Kozlov_I_S_25ISz
branch 'Kozlov_I_S_25ISz' set up to track 'origin/Kozlov_I_S_25ISz'.
vboxuser@10:~/Documents/Repo/Spring_2026$
```

Вывод:

Научился работать с системой контроля версий GIT.